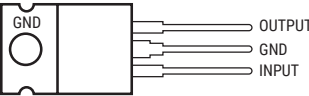


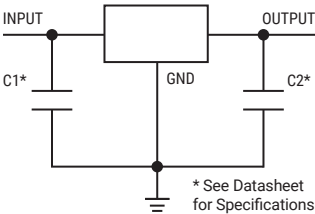
LINEAR VOLTAGE REGULATORS

Linear regulators are the simplest way to control a specific voltage with minimal external components. They are not considered to be very efficient like switching regulators. Popular part numbers shown below:

LM7805 LM7812, LM7815
TO-220 Package
5 V, 12 V, 15 V 1 A Voltage Regulators



Fixed Output Voltage Regulator



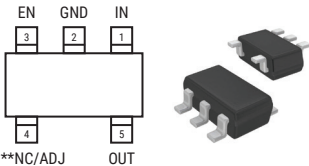
* See Datasheet for Specifications

LM78L05, LM78L12, LM78L15
TO-92 Package
5 V, 12 V, 15 V 100 mA Voltage Regulators

Pin Number	Pin Name
1	OUTPUT
2	GROUND
3	INPUT



576-2999-1-ND (Fixed)
SOT23 5 Pin
5 V 150 mA Voltage Regulator



Pin Number	Pin Name	Pin Function
1	IN	Supply Input Voltage
2	GND	Ground
3	EN (Enable)	Logic Low or Open = Shutdown
**4	NC (Fixed)	No Connect
	ADJ (Adjust)	Adjust: Connect Resistor Voltage Divider
5	OUT	Regulator Output

LDO Voltage Regulators (Low Drop Out)

These are used when the input needs to be slightly above the output. The dropout voltage is the amount of voltage above the output voltage to maintain regulation.

Popular LDO Part Numbers

Digi-Key Part Number	Input Voltage (Max)	Output Voltage	Drop Out Voltage (Max)	Package	
LP2985IM5X-3.3/NOPBCT-ND	16 V	3.3 V	580 mV @ 150 mA	SOT-23-5	
LP2985IM5-5.0/NOPBCT-ND	16 V	5 V	580 mV @ 150 mA	SOT-23-5	
LM2940IMP-12/NOPBCT-ND	16 V	12 V	1 V @ 1 A	SOT-223-4	

Switching Regulators

These regulators use a switching element to transform the incoming voltage into a pulsed voltage output which is then smoothed using capacitors, inductors, and resistors. Popular series part numbers: LM2576 – 1.23 V, 3.3 V, 5 V, 12 V, 15 V; Adjustable Output Versions in TO220-5, TO263-5 and D2PAK-5 (SMD)

DC/DC Converters

These change the input DC voltage to another voltage either lower (Buck) or higher (Boost) at the output. Some popular part numbers:

Part Number	Type	Input Voltage	Output Voltage	Current Output	Package
LM2675MX-5.0	Buck	6.5 V (min) - 40 V (max)	5 V	1 A	8-SOIC
LM2675N-5.0	Buck	6.5 V (min) - 40 V (max)	5 V	1 A	8-PDIP
LTC3525DISC6-3.3#	Boost	0.5 V (min)	3.3 V	140 mA	SC70-6
TPS61222DCKR	Boost	0.7 V (min)	5 V	200 mA	SC70-6